

OUTPUT

Q1.

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QUESTION 1 - RAIN AND WET GROUND
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RESOLUTION REFUTATION
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Initial CNF Clauses:
C1 =  $W \vee \sim R$ 
C2 =  $R$ 
C3 =  $\sim W$ 

Negated Goal:
 $\sim(W)$  =  $\sim W$ 

Resolution Process
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Iteration 1

Resolve:  $(W \vee \sim R)$  and  $(R)$ 
Resolvent =  $W$ 

Resolve:  $(W \vee \sim R)$  and  $(\sim W)$ 
Resolvent =  $\sim R$ 

Iteration 2
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Resolve:  $(W \vee \sim R)$  and  $(R)$ 
Resolvent =  $W$ 

Resolve:  $(W \vee \sim R)$  and  $(\sim W)$ 
Resolvent =  $\sim R$ 

Resolve:  $(R)$  and  $(\sim R)$ 
Resolvent =  $\square$ 
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EMPTY CLAUSE DERIVED:  $\square$ 
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Goal:  $W$ 
Conclusion: GOAL IS PROVED
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Q2.

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QUESTION 2 - STUDENT ASSIGNMENT SUBMISSION  
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RESOLUTION REFUTATION

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Initial CNF Clauses:

C1 = $M \vee \sim S$

C2 = S

C3 = $\sim M$

Negated Goal:

$\sim(M) = \sim M$

Resolution Process

Iteration 1

Resolve: $(M \vee \sim S)$ and (S)

Resolvent = M

Resolve: $(M \vee \sim S)$ and $(\sim M)$

Resolvent = $\sim S$

Iteration 2

Resolve: $(M \vee \sim S)$ and (S)

Resolvent = M

Resolve: $(M \vee \sim S)$ and $(\sim M)$

Resolvent = $\sim S$

Resolve: (S) and $(\sim S)$

Resolvent = $\square$

EMPTY CLAUSE DERIVED: $\square$

Goal: M

Conclusion: GOAL IS PROVED

Q3.

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QUESTION 3 - LIBRARY MEMBERSHIP
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RESOLUTION REFUTATION
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Initial CNF Clauses:
C1 =  $B \vee \sim L$ 
C2 =  $L$ 
C3 =  $\sim B$ 

Negated Goal:
 $\sim(B) = \sim B$ 

Resolution Process
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Iteration 1

Resolve:  $(B \vee \sim L)$  and  $(L)$ 
Resolvent =  $B$ 

Resolve:  $(B \vee \sim L)$  and  $(\sim B)$ 
Resolvent =  $\sim L$ 

Iteration 2

Resolve:  $(B \vee \sim L)$  and  $(L)$ 
Resolvent =  $B$ 

Resolve:  $(B \vee \sim L)$  and  $(\sim B)$ 
Resolvent =  $\sim L$ 

Resolve:  $(L)$  and  $(\sim L)$ 
Resolvent =  $\square$ 
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Q4.

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QUESTION 4 - PLACEMENT ELIGIBILITY
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RESOLUTION REFUTATION
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Initial CNF Clauses:
C1 =  $E \vee \sim A$ 
C2 =  $A$ 
C3 =  $\sim E$ 

Negated Goal:
 $\sim(E) = \sim E$ 

Resolution Process
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Iteration 1

Resolve:  $(E \vee \sim A)$  and  $(A)$ 
Resolvent =  $E$ 

Resolve:  $(E \vee \sim A)$  and  $(\sim E)$ 
Resolvent =  $\sim A$ 

Iteration 2

Resolve:  $(E \vee \sim A)$  and  $(A)$ 
Resolvent =  $E$ 

Resolve:  $(E \vee \sim A)$  and  $(\sim E)$ 
Resolvent =  $\sim A$ 

Resolve:  $(A)$  and  $(\sim A)$ 
Resolvent =  $\square$ 

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EMPTY CLAUSE DERIVED:  $\square$ 
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Goal:  $E$ 
Conclusion: GOAL IS PROVED
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Q5.

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QUESTION 5 - ACCESS CONTROL SYSTEM
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RESOLUTION REFUTATION
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Initial CNF Clauses:

$C1 = A \vee \sim P$

$C2 = G \vee \sim A$

$C3 = P$

$C4 = \sim G$

Negated Goal:

$\sim(G) = \sim G$

Resolution Process

Iteration 1

Resolve: $(A \vee \sim P)$ and $(G \vee \sim A)$

Resolvent = $G \vee \sim P$

Resolve: $(A \vee \sim P)$ and (P)

Resolvent = A

Resolve: $(G \vee \sim A)$ and $(\sim G)$
Resolvent = $\sim A$

Iteration 2

Resolve: $(A \vee \sim P)$ and $(G \vee \sim A)$
Resolvent = $G \vee \sim P$

Resolve: $(A \vee \sim P)$ and (P)
Resolvent = A

Resolve: $(A \vee \sim P)$ and $(\sim A)$
Resolvent = $\sim P$

Resolve: $(G \vee \sim A)$ and $(\sim G)$
Resolvent = $\sim A$

Resolve: $(G \vee \sim A)$ and (A)
Resolvent = G

Resolve: (P) and $(G \vee \sim P)$
Resolvent = G

Resolve: $(\sim G)$ and $(G \vee \sim P)$
Resolvent = $\sim P$

Resolve: (A) and $(\sim A)$
Resolvent = $\square$